Unit 1.8 Set Language and Notation

Paper 1: Basic

1. (a) $\xi = \{a, e, g, i, n, o, p, q, r, s, u\}$

   $A = \{s, i, n, g, a, p, o, r, e\}$

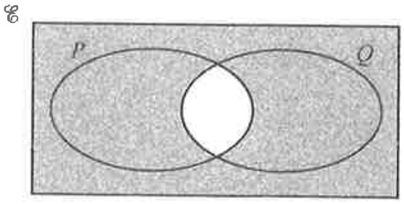
   $B = \{s, q, u, a, r, e\}$

   Find

   (i) $A \cap B$, [1]

   (ii) $(A \cup B)'$. [1]

   (b) Use set notation to describe the shaded region.



[1]

[N20/I/16]

2. $\xi = \{\text{integers } x : 4 < x < 16\}$

   $P = \{\text{prime numbers}\}$

   $Q = \{\text{multiples of } 4\}$

   $R = \{\text{factors of } 30\}$

   (a) List the elements in

   (i) R, [1]

   (ii) $P \cap R$. [1]

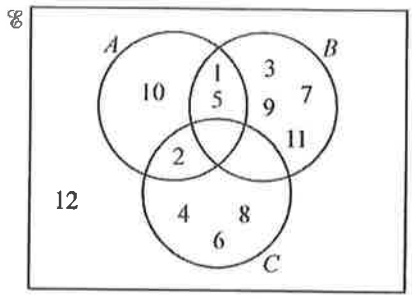
   (b) Underline the correct statements from the list below.

   $R \cap Q = \varnothing$ $\quad Q \cup R = \{5, 6, 8, 10, 12\}$ $\quad P \cap Q = \{0\}$ $\quad Q \subset R$ $\quad 16 \notin P$ [2]

[N19/I/18]

3. $\xi = \{\text{integers } x : 1 \leqslant x \leqslant 12\}$

   The Venn diagram shows the elements of ξ and three sets A, B and C.



Use one of the symbols below to complete each statement.

$\varnothing \quad \subset \quad \not\subset \quad \notin \quad \in \quad \xi$

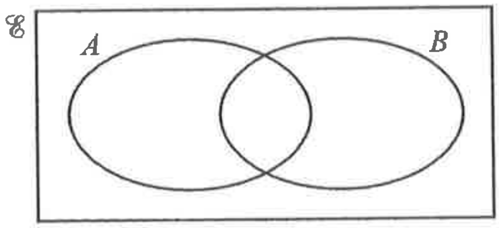
(a) $\{2, 10\}$ .... A [1]

(b) 3 .... B [1]

(c) $B \cap C =$ .... [1]

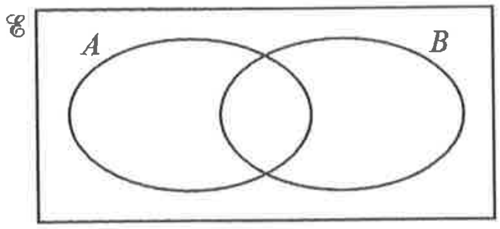
[N17/I/6]

4. **(a)** On the Venn diagram, shade the region which represents $A \cup B'$.



[1]

(b) On the Venn diagram, shade the region which represents $A' \cap B'$.



[1]

[N16/I/3]

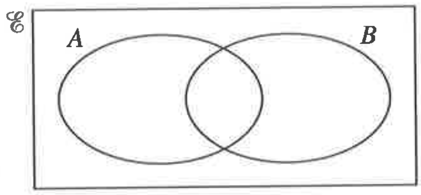
5. $\xi = \{\text{integers } x : 1 \leqslant x \leqslant 8\}$

$A = \{\text{factors of } 6\}$

$B = \{\text{prime numbers}\}$

(a) List the elements in $A' \cap B'$. [1]

(b) On the Venn diagram, shade the region which represents $A' \cap B'$.

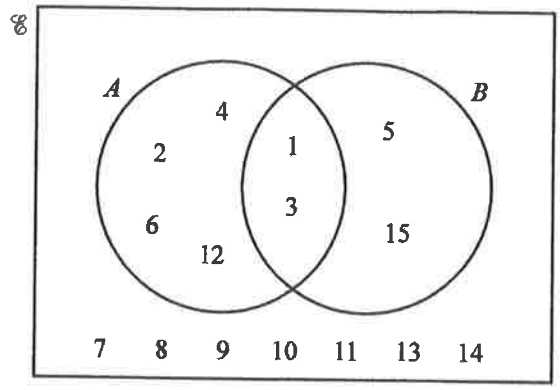


[1]

[N15/I/6]

6. $\xi = \{\text{integers } x : 1 \leqslant x \leqslant 15\}$
   $A = \{1, 2, 3, 4, 6, 12\}$
   $B = \{1, 3, 5, 15\}$

   This information is shown on the Venn diagram.

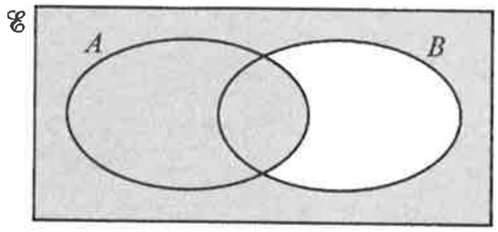


   (a) Describe the elements of set A. [1]
   (b) List the elements contained in the set $A' \cap B'$. [1]
   (c) Find the number of elements in $(A \cap B') \cup (A' \cap B)$. [1]

[N21/I/12]

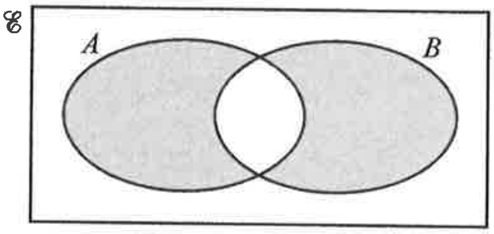
7. Write down the sets represented by the following shaded regions.

   (a)



[1]

   (b)



[1]

[N18/I/7]

Paper 2: Basic

1. $\xi = \{\text{integers } x : 1 \leqslant x \leqslant 15\}$

   $A = \{\text{prime numbers}\}$

   $B = \{\text{factors of } 30\}$

   $C = \{\text{multiples of } 3\}$

   (i) List the elements in A'. [1]

   (ii) List the elements in $(A \cup B)'$. [1]

   (iii) A number, p, is chosen at random from the set $(B \cup C)$.
   Find the probability that $p \notin C$. [2]

   [N22/II/9(a)]